MILITARY SPECIFICATION

MICROCIRCUITS, MEMORY, DIGITAL, CMOS 524,288 BIT, ELECTRICALLY ERASABLE, PROGRAMMABLE READ-ONLY MEMORY (EEPROM), MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, CMOS, 64K words x 8 bit, electrically clearable, writable read-only memory microcircuits. Two product assurance classes (S and B), a choice of lead finish and two package types are provided for each device and are reflected in the complete Part Number or Identifying Number (PIN).

1.2 Part or Identifying Number (PIN). The PIN shall be in accordance with MIL-M-38510.

1.2.1 Device types. The device types shall be as follows:

Device type	Circuit organization	Access time	Endurance
01	64K words x 8 bit	300 ns	100 cycles
02	64K words x 8 bit	300 ns	1,000 cycles
03	64K words x 8 bit	250 ns	100 cycles
04	64K words x 8 bit	250 ns	1,000 cycles
05	64K words x 8 bit	200 ns	100 cycles
06	64K words x 8 bit	200 ns	1,000 cycles
07	64K words x 8 bit	150 ns	100 cycles
08	64K words x 8 bit	150 ns	1,000 cycles
09	64K words x 8 bit	120 ns	100 cycles
10	64K words x 8 bit	120 ns	1,000 cycles

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outlines. The case outlines shall be designated as follows:

Letter	Case outline (see MIL-M-38510, appendix C)
X	See figure 1 (32-lead, 1.685" x .600" x .225"), dual-in-line package
Y	C-12 (32-terminal, .560" x .458" x .120"), rectangular chip carrier package
Z	See figure 1 (32-lead, .830" x .416" x .120"), flat package

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addresssed to: Rome Air Development Center, RBE-2, Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5962

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1.3 Absolute maximum ratings. 1/

Supply voltage range (V_{CC})	-0.3 V dc to +6.0 V dc
Storage temperature range	-65°C to +150°C
Maximum power dissipation (P_D)	+1.0 W
Lead temperature (soldering, 10 seconds)	+300°C
Junction temperature (T_J) 2/	+175°C
Thermal resistance, junction-to-case (θ_{JC}) 3/	
Case X	21°C/W
Cases Y and Z	18°C/W
Input voltage range (V_{IL}, V_{IH})	-0.5 V dc to V_{CC} 1.0
Endurance (minimum of)	
Device types 01, 03, 05, 07, and 09	100 cycles/byte
Device types 02, 04, 06, 08, and 10	1,000 cycles/byte
Program/erase voltage (Vpp)	14.0 V dc
Data retention	10 years (minimum)

1.4 Recommended operating conditions.

Supply voltage range (V_{CC})	+4.5 V dc to +5.5 V dc
Case operating temperature range (T_C)	-55°C to +125°C
Input voltage, low (V_{IL})	-0.3 V dc to +0.8 V dc
Input voltage, high (V_{IH})	+2.0 V dc to V_{CC} +0.3 V
Chip clear voltage (V_P)	11.4 V dc to 12.6 V dc

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.3).

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for Microelectronics.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standard Document Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

1/ All voltages referenced to V_{SS} (ground).
2/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions in accordance with method 5004 of MIL-STD-883.
3/ When a thermal resistance is included in MIL-M-38510, appendix C, it shall supersede the value stated herein.

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2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specifications. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Terminal connections. The terminal connections shall be as specified on figure 2.

3.2.2 Truth table. The mode selection shall be as specified on figure 3.

3.2.3 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.5).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full case operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups of table II. The electrical tests for each subgroup are described in table I.

3.6 Marking. Marking shall be in accordance with MIL-M-38510 and 1.2 herein. At the option of the manufacturer, the country of origin may be omitted from the body of the microcircuit but shall be retained on the initial container.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 42 (see MIL-M-38510, appendix E).

3.8 Processing of flash EEPROMs. All testing required and quality assurance provisions herein shall be satisfied by the manufacturer prior to delivery.

3.8.1 Conditions of the supplied devices. Devices will be supplied in a written or cleared state. No provision will be made for supplying programmed devices.

3.8.2 Clearing of flash EEPROMs. When specified, devices shall be cleared in accordance with the procedures and characteristics specified in 4.5.3.

3.8.3 Writing of flash EEPROMs. When specified, devices shall be written in accordance with the procedures and characteristics specified in 4.5.3.

3.8.4 Verification of state of flash EEPROMs. When specified, devices shall be verified as either written to the specified pattern or cleared. As a minimum, verification shall consist of performing a read of the entire array to verify that all bits are in the proper state. Any bit that does not verify to be in the proper state shall constitute a device failure and the device shall be removed from the lot or sample.

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TABLE I. Electrical performance characteristics.

Characteristic	Symbol	Conditions $-55°C \leq T_C \leq +125°C$ $4.5\ V \leq V_{CC} \leq 5.5\ V$	Device type	Subgroups	Limits		Unit
					Min	Max	
Parametric characteristics 1/ 2/							
Supply current (active)	I_{CC1}	$\overline{CE} = V_{IL}$ All I/O's = 0 mA, Inputs = 5 MHz at V_{IL} to V_{IH} V_{CC} = 5.5 V	All	1, 2, 3		60	mA
Supply current (TTL standby)	I_{CC2}	$\overline{CE} = V_{IH}$, $\overline{OE} = V_{IL}$, All I/O's = 0 mA, Inputs = V_{IH} V_{CC} = 5.5 V	All	1, 2, 3		5	mA
Supply current (CMOS standby)	I_{CC3}	$\overline{CE} = V_{CC} - 0.3$ $\overline{OE}$ = 0.3 V All I/O's = 0 mA Inputs = V_{CC} -0.3 V	All	1, 2, 3		400	µA
Input leakage (high)	I_{LIH}	V_{CC} = 5.5 V V_{IN} = 3/	All	1, 2, 3	-1	1	µA
Input leakage (low)	I_{LIL}	V_{CC} = 5.5 V V_{IN} = 4/	All	1, 2, 3	-1	1	µA
Output leakage (high)	I_{LOH}	V_{CC} = 5.5 V V_{OUT} = 5.5 V 5/	All	1, 2, 3	-10	10	µA
Output leakage (low)	I_{LOL}	V_{CC} = 5.5 V V_{OUT} = 0.1 V 5/	All	1, 2, 3	-10	10	µA
Input voltage low	V_{IL}	V_{CC} = 5.5 V	All	1, 2, 3	-0.3	0.8	V
Input voltage high	V_{IH}	V_{CC} = 4.5 V	All	1, 2, 3	2.0	V_{CC} +1.0	V
Output voltage low	V_{OL}	V_{CC} = 4.5 V 6/ I_{OL} = 2.1 mA	All	1, 2, 3		0.40	V
Output voltage high	V_{OH}	V_{CC} = 4.5 V 7/ I_{OL} = -400 µA	All	1, 2, 3	2.4		V

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Characteristic	Symbol	Conditions $-55°C \le T_C \le +125°C$ $4.5\ V \le V_{CC} \le 5.5\ V$	Device type	Subgroups	Limits		Unit
					Min	Max	
Dynamic characteristics							
Input capacitance 2/ 8/	C_I	V_I = 0 V, f = 1 MHz	All	4		6	pF
Output capacitance 2/ 8/	C_O	V_O = 0 V, f = 1 MHz	All	4		12	pF
Write/clear voltage	V_P		All	9, 10, 11	11.4	12.6	V
Vpp voltage during read	V_{PRP}		All	9, 10, 11	0	V_P	V
Output level (TTL)	V_{OH1}	I_{OH} = -400 µA	All	9, 10, 11	2.4		V
Output level (CMOS)	V_{OH2}	I_{OH} = -100 µA	All	9, 10, 11	V_{CC} -1.0		V
Vpp current standby mode	I_{PS}	CE = V_{IL}, Vpp = V_P	All	9, 10, 11		200	µA
Vpp current read mode	I_{PR}	CE = V_{IL}, Vpp = V_P	All	9, 10, 11		200	µA
Vpp current byte write mode	I_{PP}	Vpp = V_P	All	9, 10, 11		40	mA
Vpp current chip clear	I_{PE}	Vpp = V_P	All	9, 10, 11		100	mA
Vpp current sector clear	I_{PE}	Vpp = V_P	All	9, 10, 11		25	mA
Switching characteristics, read operation 2/ 9/							
Read cycle time 10/	t_{AVAV}	See figures 4 and 5	01, 02	9, 10, 11	300		ns
			03, 04		250		ns
			05, 06		200		ns
			07, 08		150		ns
			09, 10		120		ns
Address access time	t_{AVQV}		01, 02	9, 10, 11		300	ns
			03, 04			250	ns
			05, 06			200	ns
			07, 08			150	ns
			09, 10			120	ns

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Characteristic	Symbol	Conditions $-55°C \le T_C \le +125°C$ $4.5\ V \le V_{CC} \le 5.5\ V$	Device type	Subgroups	Limits		Unit
					Min	Max	
Switching characteristics, read operation 2/ 9/							
Chip enable to data	t_{ELQV}	See figures 4 and 5	01, 02	9, 10, 11		300	ns
			03, 04			250	ns
			05, 06			200	ns
			07, 08			150	ns
			09, 10			120	ns
Output enable to data	t_{OLQV}		01, 02	9, 10, 11		150	ns
			03, 04			100	ns
			05, 06			75	ns
			07, 08			65	ns
			09, 10			60	ns
Output hold time 8/	t_{OHQX}		All	9, 10, 11	0		ns
$\overline{OE}/\overline{CE}$ to data float 8/	t_{EHQZ} t_{OHQZ}		01, 02	9, 10, 11		100	ns
			03, 04			60	ns
			05, 06			50	ns
			07, 08			45	ns
			09, 10			40	ns
Output enable to output in low Z 8/	t_{OLQX} t_{ELQX}		All	9, 10, 11	0		ns
Switching characteristics, write cycle operation 10/							
Write cycle time 10/	t_{WHWL} t_{EHEL}	See figures 4 and 5	All	9, 10, 11	75		µs
Address setup time 10/	t_{AVWL} t_{AVEL}		All	9, 10, 11	20		ns
Address hold time 10/	t_{WLAX} t_{ELAX}		All	9, 10, 11	100		ns
Write setup time 10/	t_{ELWL} t_{WLEL}		All	9, 10, 11	0		ns
Write hold time 10/	t_{WHEH} t_{EHWH}		All	9, 10, 11	0		ns
Output enable setup time 10/	t_{OHWL} t_{OHEL}		All	9, 10, 11	10		ns

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Characteristic	Symbol	Conditions $-55°C \le T_C \le +125°C$, $4.5\ V \le V_{CC} \le 5.5\ V$	Device type	Subgroups	Limits		Unit
					Min	Max	
Switching characteristics, read operation 2/ 9/							
Output enable hold time 10/	t_{WHOL} t_{EHOL}	See figures 4 and 5	All	9, 10, 11	10		ns
Write pulse width 10/	t_{WLWH} t_{ELEH}		All	9, 10, 11	100		ns
Data setup time 10/	t_{DVWH} t_{DVEH}		All	9, 10, 11	50		ns
Data hold time 10/	t_{WHDX} t_{EHDX}		All	9, 10, 11	10		ns
Vpp setup time 10/	t_{PVpWL} t_{PVpEL}		All	9, 10, 11	2		µs
Vpp hold time 10/	t_{WHPX} t_{EHPX}		All	9, 10, 11	150		µs
Write recover time 10/	t_{WHEL} t_{EHWL}		All	9, 10, 11		1.5	ms
Switching characteristics, sector clear							
$\overline{CE}$ setup time 10/	t_{ELWL}	See figures 4 and 5	All	9, 10, 11	0		ns
$\overline{OE}$ setup time 10/	t_{OHWL}		All	9, 10, 11	0		ns
$\overline{WE}$ pulse width 10/	t_{WLWH}		All	9, 10, 11	100		ns
$\overline{CE}$ hold time 10/	t_{WHEH}		All	9, 10, 11	0		ns
$\overline{OE}$ hold time 10/	t_{WHOH}		All	9, 10, 11	0		ns
Clear time 10/	t_{WHWL}		All	9, 10, 11	500		ms

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Characteristic	Symbol	Conditions $-55°C \leq T_C \leq +125°C$ $4.5\ V \leq V_{CC} \leq 5.5\ V$	Device type	Subgroups	Limits		Unit
					Min	Max	
Switching characteristics, sector clear							
High voltage 10/	V_P	See figures 4 and 5	All	9, 10, 11	11.4	12.6	V
Vpp setup time 10/	t_{PVpWL}		All	9, 10, 11	2		µs
Vpp hold time 10/	t_{WHPX}		All	9, 10, 11	500		ms
Data setup time 10/	t_{DVWH}		All	9, 10, 11	50		ns
Data hold time 10/	t_{WHDX}		All	9, 10, 11	10		ns
Abort time 10/	t_{EHEL} (t_{ABORT})		All	9, 10, 11		250	µs
Address setup 10/	t_{AVWL}		All	9, 10, 11	20		ns
Address hold 10/	t_{WLAX}		All	9, 10, 11	100		ns
Clear recovery time 10/	t_{VPXEL}		All	9, 10, 11		250	ms
Switching characteristics, chip erase							
$\overline{CE}$ setup time 10/	t_{ELWL}	See figures 4 and 5	All	9, 10, 11	0		ns
$\overline{OE}$ setup time 10/	t_{OHWL}		All	9, 10, 11	0		ns
$\overline{WE}$ pulse width 10/	t_{WLWH}		All	9, 10, 11	100		ns
$\overline{CE}$ hold time 10/	t_{WHEH}		All	9, 10, 11	0		
$\overline{OE}$ hold time 10/	t_{WHOH}		All	9, 10, 11	0		
Clear time 10/	t_{WHWH}		All	9, 10, 11	500		ms

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Characteristic	Symbol	Conditions $-55°C \leq T_C \leq +125°C$ $4.5\ V \leq V_{CC} \leq 5.5\ V$	Device type	Subgroups	Limits		Unit
					Min	Max	
High voltage 10/	V_P	See figures 4 and 5	All	9, 10, 11	11.4	13.0	V
V_{PP} setup time 10/	t_{PVPWL}		All	9, 10, 11	2		ms
V_{PP} hold time 10/	t_{WHPX}		All	9, 10, 11	500		ms
Data setup time 10/	t_{DVWH}		All	9, 10, 11	50		ns
Data hold time 10/	t_{WHDX}		All	9, 10, 11	10		ns
Clear recovery 10/	t_{VPXEL}		All	9, 10, 11		250	ms

1/ DC and read mode.

2/ All pins not being tested are to be open.

3/ V_{IN} = 5.5 V for pin being tested, all other pins not tested at 0.1 V (except I/O pins which are open).

4/ V_{IN} = 0.1 V for pin being tested, all other pins not tested at 5.5 V (except I/O pins which are open).

5/ Connect all address inputs and $\overline{OE}$ to V_{IH} and measure I_{LOL} and I_{LOH} with the output under test connected to V_{OUT}.

6/ An input preconditioning logic sequence shall be applied that results in a logic "0" at the output to be measured. Logic input levels are V_{IL} = 0.8 V and 2.0 V.

7/ An input preconditioning logic sequence shall be applied that results in a logic "1" at the output to be measured. Logic input levels are V_{IL} = 0.8 V and 2.0 V.

8/ Only performed for initial qualification and after any design or process change that could affect that parameter and therefore, guaranteed to the limits specified in table I.

9/ Equivalent ac test conditions: Output load: 1 TTL gate and C_1 = 100 pF, input rise and and fall times $\leq$ 10 ns, input pulse levels: 0.40 and 2.4 V.
Timing measurement reference levels:
Inputs 1 V and 2 V
Outputs 0.8 V and 2.4 V.

10/ Tested by application of specified timing signals and conditions and therefore guaranteed to limits specified in table I.

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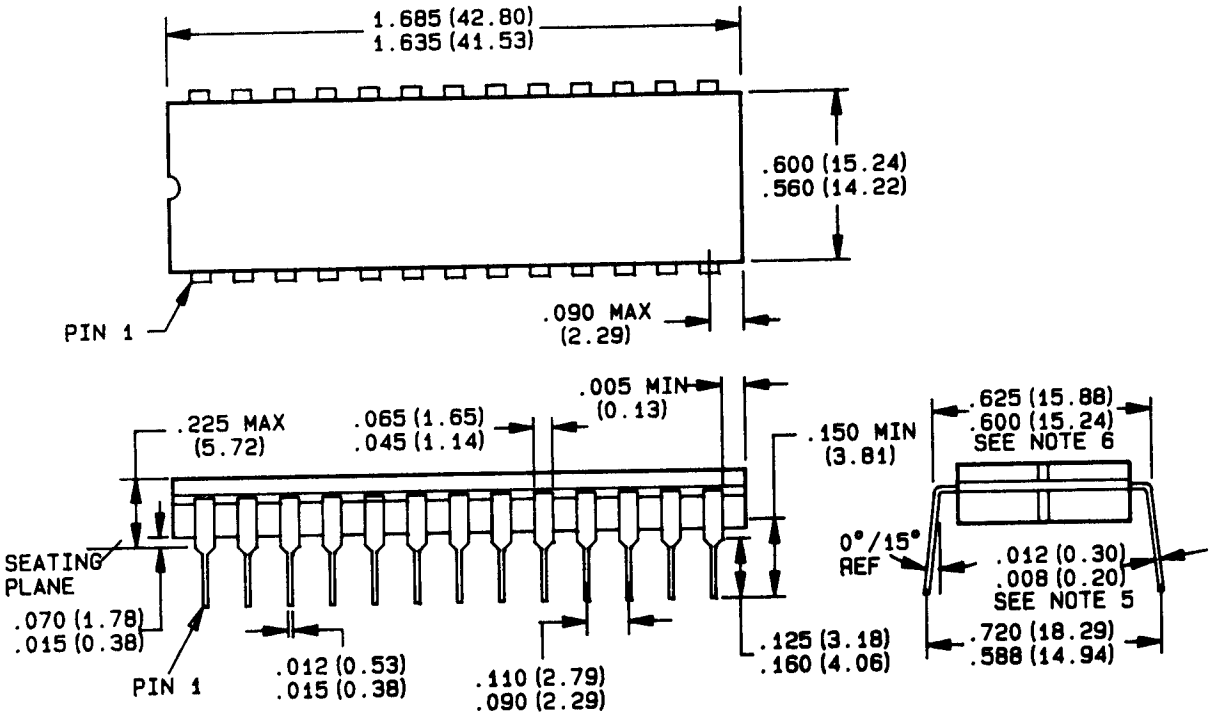
TABLE II. Electrical test requirements.

Line number	MIL-STD-883 test requirements	Class S device 1/ 2/ 3/ 4/			Class B device 1/ 2/ 3/ 4/		
		Reference paragraph	Table I subgroups 5/	Table III delta limits 6/	Reference paragraph	Table I subgroups 5/	Table III delta limits 6/
1	Interim electrical parameters (method 5004)		1,7,9			1,7,9 or 2,8A,10	
2	Static burn-in (method 1015)	4.2c 4.5.2	Required			Not required	
3	Same as line 1	4.5.3	1*	Δ 7/			
4	Dynamic burn-in (method 1015)	4.2b 4.5.2	Required		4.2b 4.5.2	Required	
5	Same as line 1		1*	Δ 7/			
6	Final electrical parameters (method 5004)	4.5.3	1*,2,3,7, 8,9,10,11			1*,2,3,7, 8,9,10,11	
7	Group A test requirements (method 5005)	4.4.1	1,2,3,4,7, 8,9,10,11		4.4.1	1,2,3,4,7, 8,9,10,11	
8	Group B end-point electrical parameters (method 5005)	4.4.2 4.5.3	1,2,3,7, 8,9,10,11	Δ	4.4.2		
9	Group C end-point electrical parameters (method 5005)				4.4.3 4.5.3	1,2,3,7, 8,9,10,11	Δ
10	Group D end-point electrical parameters (method 5005)	4.4.4	1,2,3,7, 8,9,10,11		4.4.4	1,2,3,7, 8,9,10,11	

1/ Blank spaces indicate tests are not applicable.
2/ For all electrical tests, the device shall be programmed to the data pattern specified.
3/ Any or all subgroups at the same temperature may be combined when using a multifunction tester.
4/ Subgroups 7 and 8 shall consist of writing or reading the data patterns specified in accordance with the limits of table I, subgroups 9, 10, and 11.
5/ * indicates PDA applies to subgroup 1 (see 4.2.1).
6/ Δ indicates delta limit (see table III) shall be required where specified, and the delta values shall be computed with reference to the previous interim electrical parameters (line 1).
7/ The device manufacturer may at his option, either complete subgroup 1 electrical parameter measurements, including delta measurements, within 96 hours after burn-in completion (removal of bias); or may complete subgroup 1 electrical measurements without delta measurements within 24 hours after burn-in completion (removal of bias).

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Case X

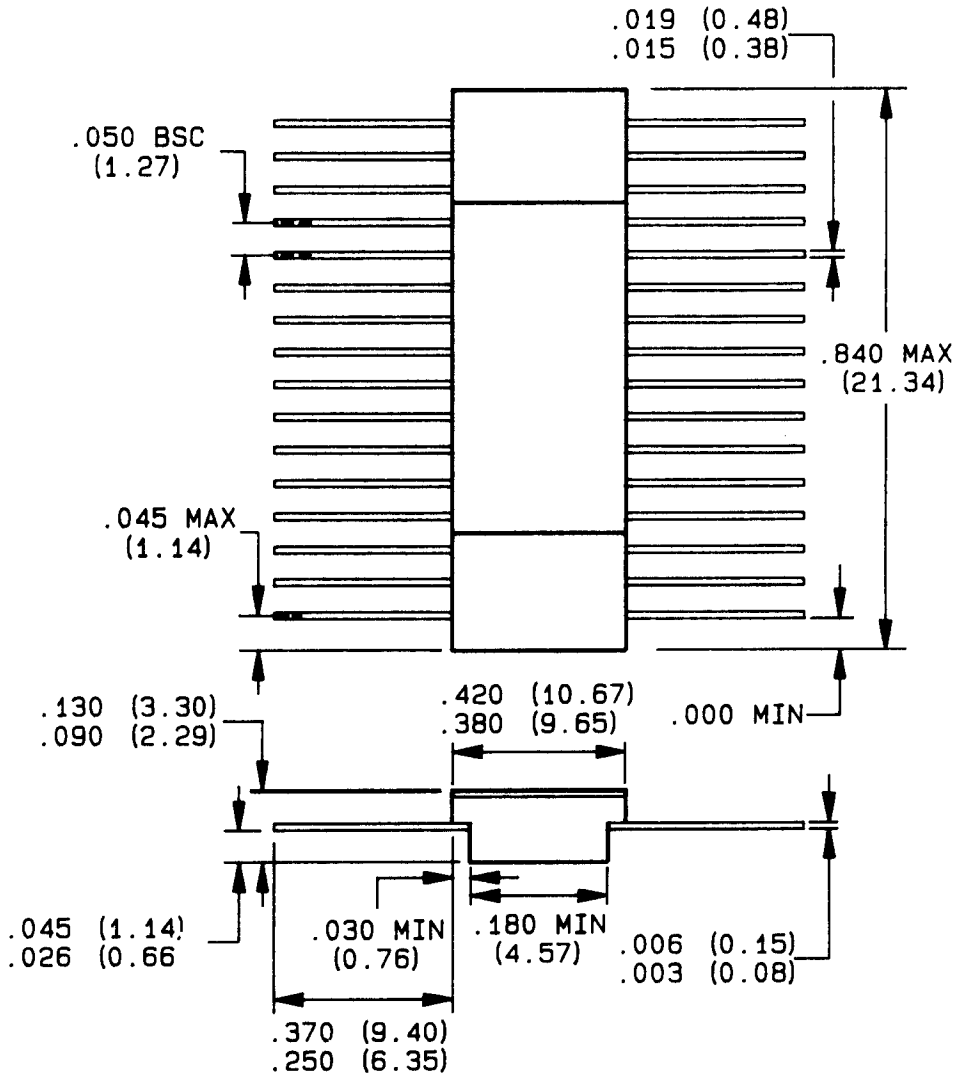


NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Millimeters are in parentheses.
4. Unless otherwise specified, tolerance is ±.005 inch (0.13 mm).
5. For solder dipped leads, thickness will be .020 maximum.
6. Dimension is measured from outside shoulder to shoulder.

FIGURE 1. Case outline X.

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Case Z



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Millimeters are in parentheses.
4. Unless otherwise specified, tolerance is ±.005 inch (0.13 mm).

FIGURE 1. Case outline Z.

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Device type(s)	01-10
Case outlines	X, Y, Z
Terminal number	Terminal symbol
1	V_{PP}
2	NC
3	A_{15}
4	A_{12}
5	A_7
6	A_6
7	A_5
8	A_4
9	A_3
10	A_2
11	A_1
12	A_0
13	I/O_0
14	I/O_1
15	I/O_2
16	V_{SS}
17	I/O_3
18	I/O_4
19	I/O_5
20	I/O_6
21	I/O_7
22	$\overline{CE}$
23	A_{10}
24	$\overline{OE}$
25	A_{11}
26	A_9
27	A_8
28	A_{13}
29	A_{14}
30	NC
31	$\overline{WE}$
32	V_{CC}

FIGURE 2. Terminal connections.

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Mode	Inputs				A9-15 A10-16	A08 A09	D07
	$\overline{CE}$	$\overline{OE}$	$\overline{WE}$	V_{pp}			
Read	V_{IL}	V_{IL}	V_{IH}	X	Address	Address	D_{OUT}
Write	V_{IL}	V_{IH}	V_{IL}	V_p	Address	Address	D_{IN}
Standby	V_{IH}	X	X	X	X	X	HI-Z
Chip clear select	V_{IL}	V_{IH}	V_{IL}	TTL	X	X	X
Chip clear	V_{IL}	V_{IH}	V_{IL}	V_p	X	X	'FF'
Sector clear	V_{IL}	V_{IH}	V_{IL}	V_p	Address	X	'FF'

Table definitions:

V_{IH} = High logic level

V_{IL} = Low logic level

V_p = Program/clear voltage

X = Do not care

High Z = High impedance state

D_{IN} = Data input

D_{OUT} = Data output

FIGURE 3. Truth table.

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Column address (see notes)

		0	1	2	3	4	5	6	225	226	.	.	509	510	511
R	0	AA	AA	AA	AA	AA	AA	AA	AA	AA	AA	AA	AA	AA	AA
O	1	55	55	55	55	55	55	55	55	55	55	55	55	55	55
W	2	AA	AA	AA	AA	AA	AA	AA	AA	AA	AA	AA	AA	AA	AA
	3	55	55	55	55	55	55	55	55	55	55	55	55	55	55
A	.														
D	.														
D	.														
R	125	AA	AA	AA	AA	AA	AA	AA	AA	AA	AA	AA	AA	AA	AA
E	126	55	55	55	55	55	55	55	55	55	55	55	55	55	55
S	127	AA	AA	AA	AA	AA	AA	AA	AA	AA	AA	AA	AA	AA	AA
S	128	55	55	55	55	55	55	55	55	55	55	55	55	55	55

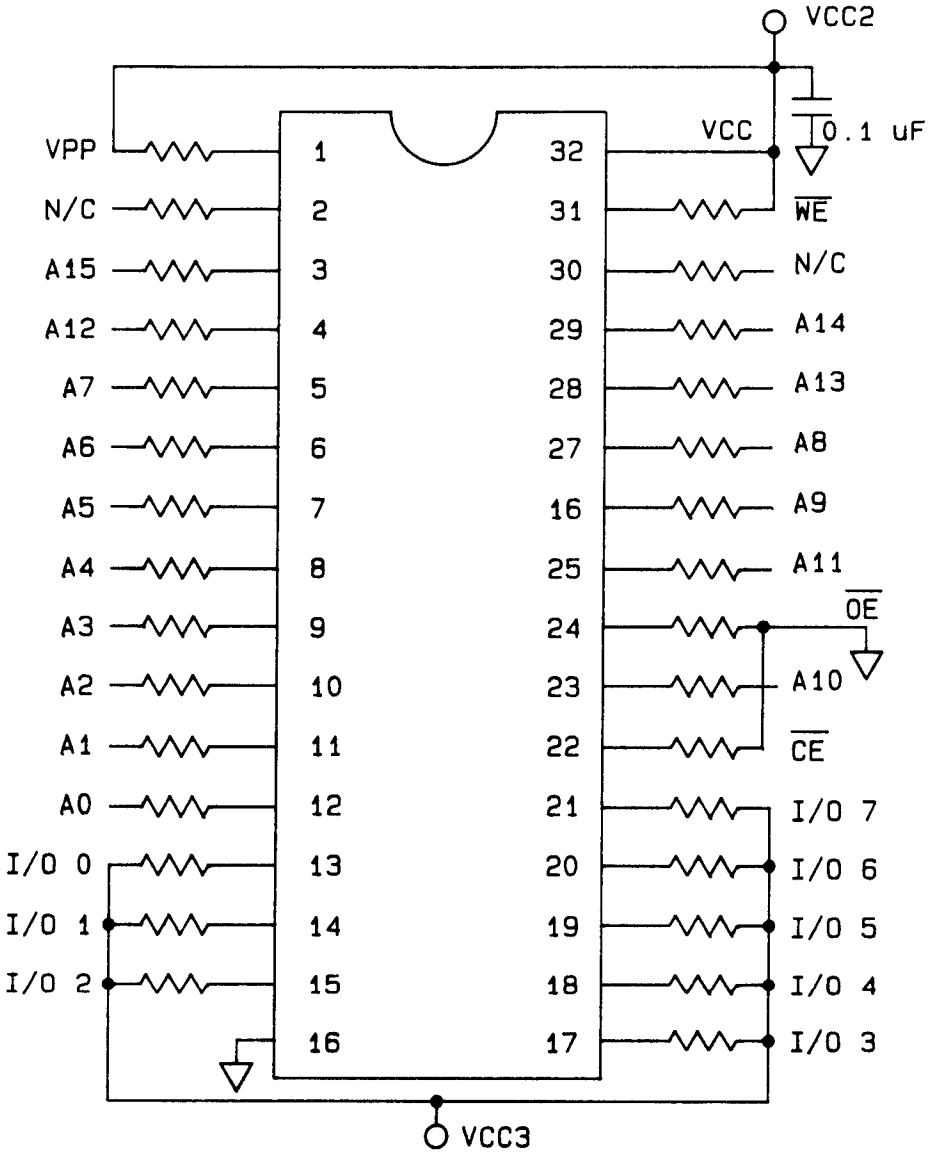
NOTES:
1. All address numbers shown in decimal.
2. Each column/row address location corresponds to 1 byte.
3. All data numbers shown in hexadecimal.
   AA = 10101010 55 = 01010101
4. Manufacturers at their option may employ an equivalent pattern provided it is a topologically true alternating bit pattern.

FIGURE 4. Data pattern.

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Cases X, Z (cerdip and flatpack)

All device types



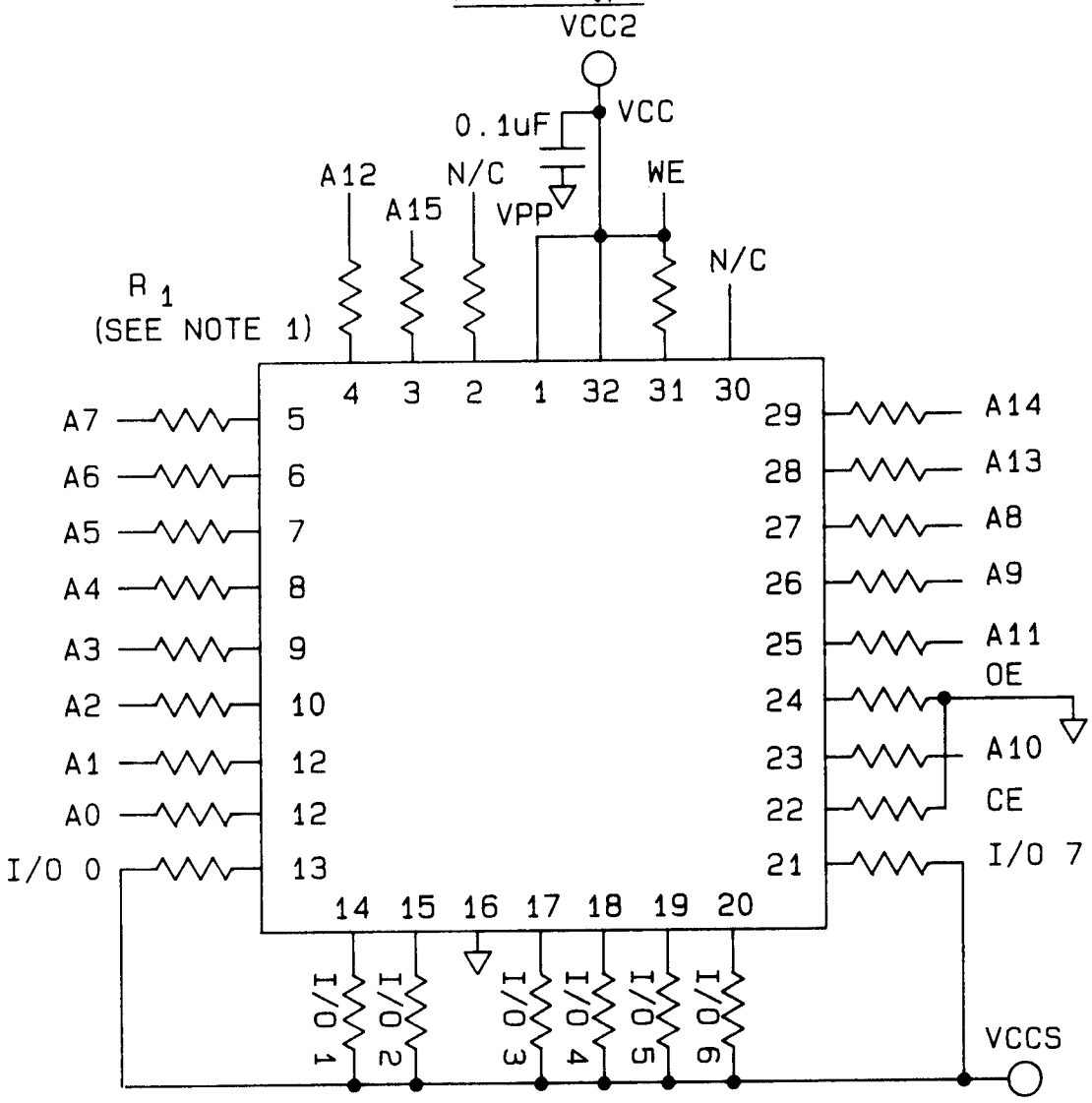
NOTES:
1. All resistors are 2.2 kΩ 0.25W, 5 percent metal film, at every socket.
2. There is a 0.1 μF decoupling capacitor between V_{CC} and GND at every socket.
3. $V_{CC1} = V_{CC2} = 5.25$ V, $V_{CC3} = 2.5$ V. All voltage levels are ±0.25 V.
4. Power up sequence: V_{CC2}, addresses, V_{CC3}.
5. Power down sequence: V_{CC3}, addresses, V_{CC2}.
6. F_0 (A_0) = 500 kHz.
7. F_1 (A_1) = F_0 divided by 2, F_2 = F_1 divided by 2 ... F_{12} (A_{12}) = F_{11} divided by 2.

FIGURE 5. Burn-in and operating life test circuit.

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Case Y (LCC)

All device types



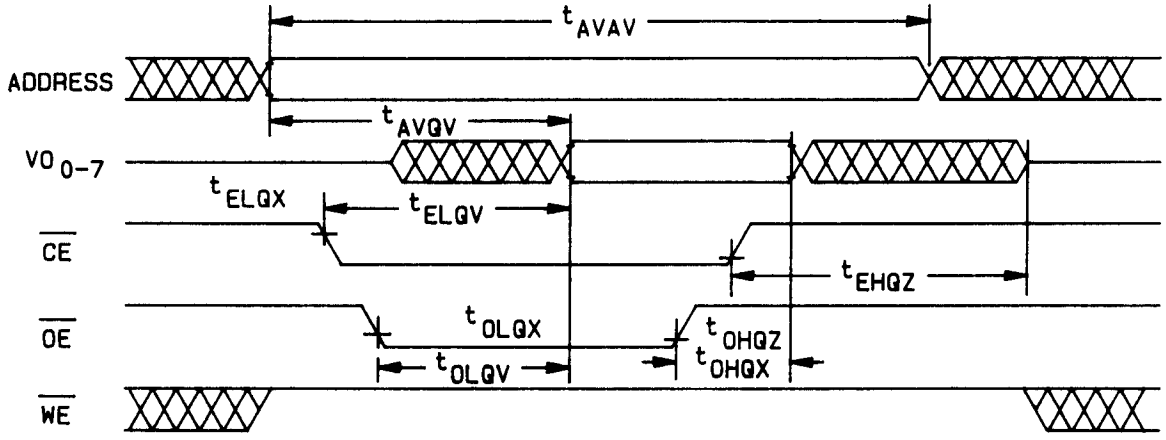
NOTES:
1. All resistors are 2.2 kΩ 0.25W, 5 percent metal film, at every socket.
2. There is a 0.1 µF decoupling capacitor between V_{CC} and GND at every socket.
3. $V_{CC1} = V_{CC2} = 5.25$ V, $V_{CC3} = 2.5$ V. All voltage levels are ±0.25 V.
4. Power up sequence: V_{CC2}, addresses, V_{CC3}.
5. Power down sequence: V_{CC3}, addresses, V_{CC2}.
6. F_0 (A_0) = 500 kHz.
7. F_1 (A_1) = F_0 divided by 2, F_2 = F_1 divided by 2 ... F_{12} (A_{12}) = F_1 divided by 2.

FIGURE 5. Burn-in and operating life test circuit - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:59Z
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Read timing waveforms

All device types



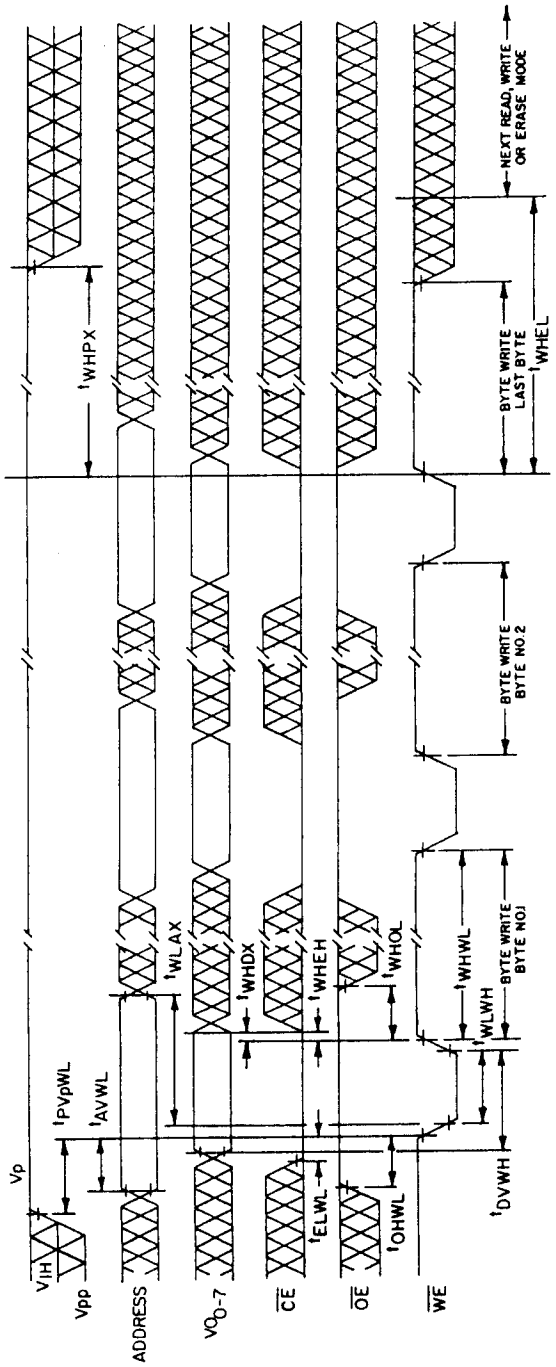
NOTE: V_{CC} shall be applied simultaneously or after $\overline{WE}$ and removed simultaneously or before $\overline{WE}$.

FIGURE 6. Switching waveforms.

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Byte write timing waveforms ($\overline{WE}$ controlled)

All device types



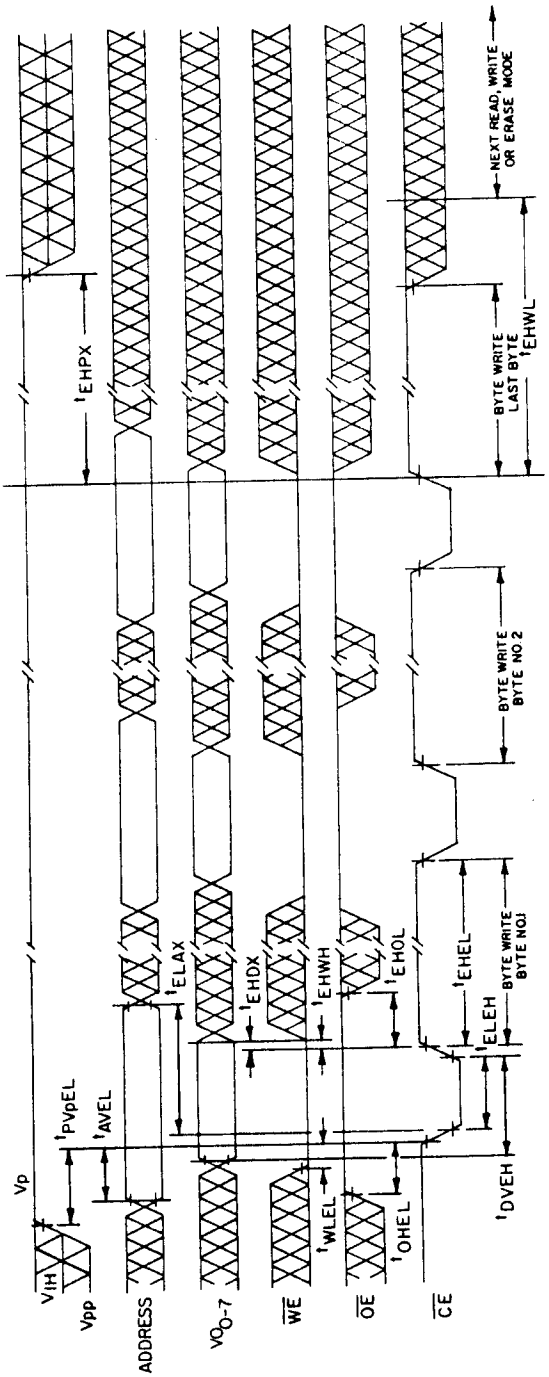
NOTE: $\overline{WE}$ and $\overline{CE}$ both must be active to initiate a write cycle; therefore, the sequence of $\overline{WE}$ or $\overline{CE}$ (e.g., for $\overline{WE}$ and $\overline{CE}$ controlled write) is verified interchangeable without duplicate testing.

FIGURE 6. Switching waveforms - Continued.

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Byte write timing waveforms ($\overline{CE}$ controlled)

All device types



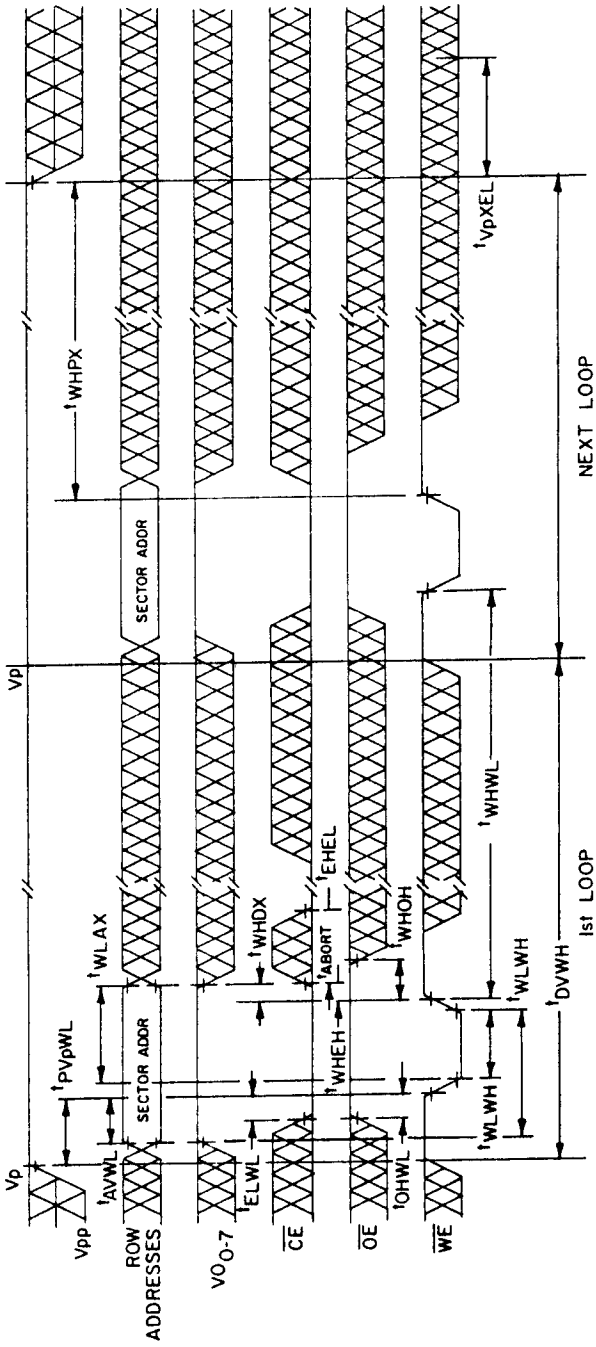
NOTE: $\overline{WE}$ and $\overline{CE}$ both must be active to initiate a write cycle; therefore, the sequence of $\overline{WE}$ or $\overline{CE}$ (e.g., for $\overline{WE}$ and $\overline{CE}$ controlled write) is verified interchangeable without duplicate testing.

FIGURE 6. Switching waveforms - Continued.

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Sector clear mode waveforms

All device types



NOTE: $\overline{WE}$ and $\overline{CE}$ both must be active to initiate a write cycle; therefore, the sequence of $\overline{WE}$ or $\overline{CE}$ (e.g., for WE and CE controlled write) is verified interchangeable without duplicate testing.

FIGURE 6. Switching waveforms - Continued.

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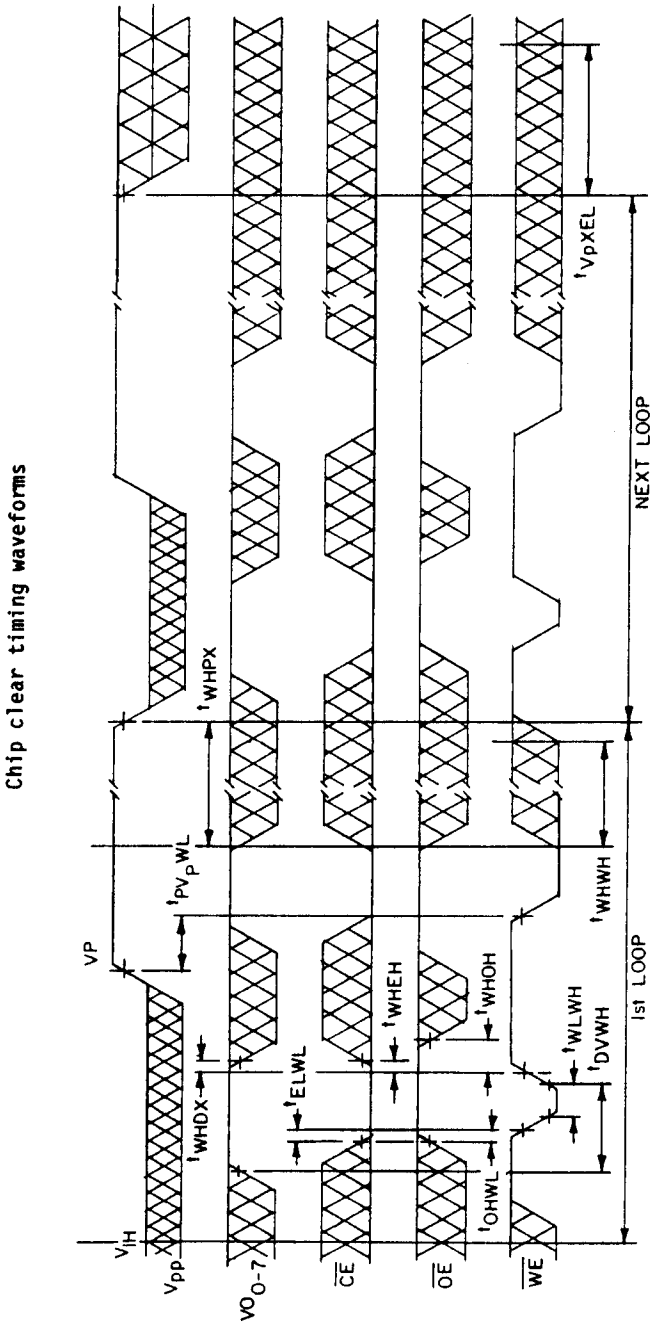
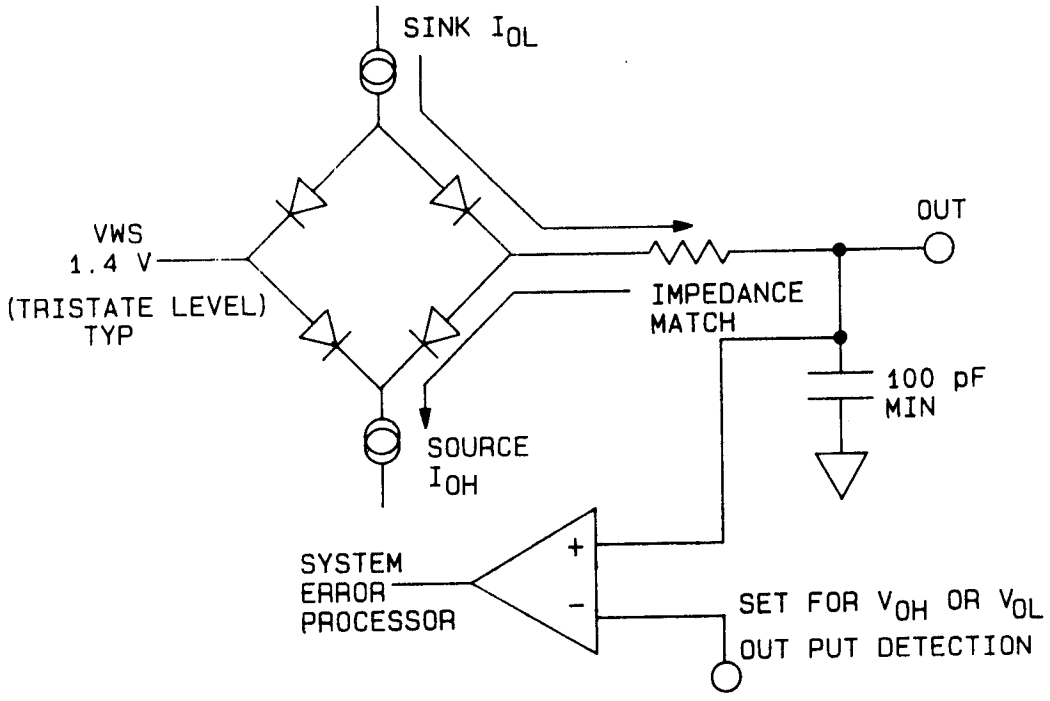


FIGURE 6. Switching waveforms - Continued.

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NOTES:
1. V_{OH1} and V_{OL0} will be adjusted to meet load conditions of table I.
2. Total capacitance within comparator, the load board, the interface plate, and the handler contractor.

FIGURE 7. Switching load circuit.

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All device types

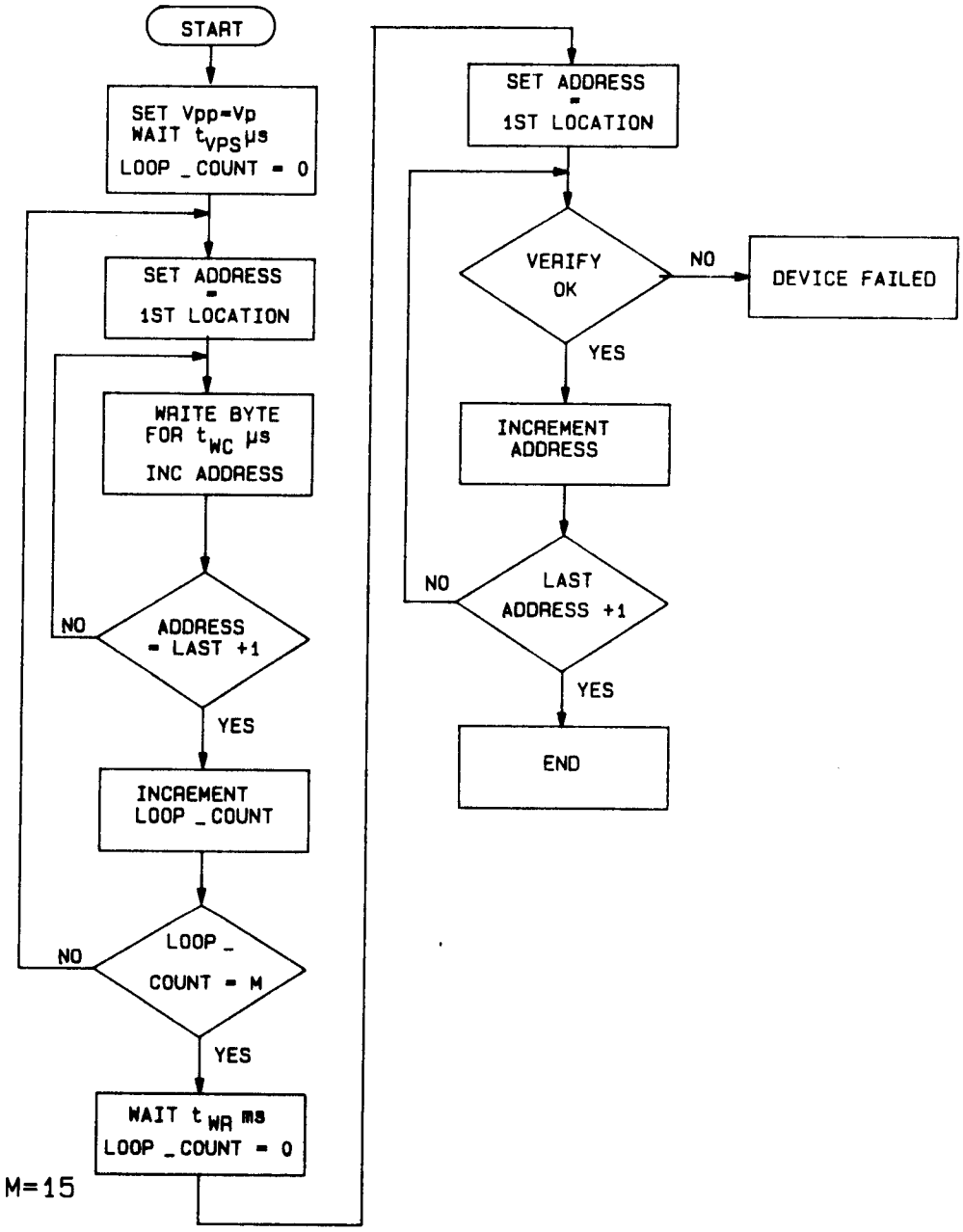


FIGURE 8. Write algorithm.

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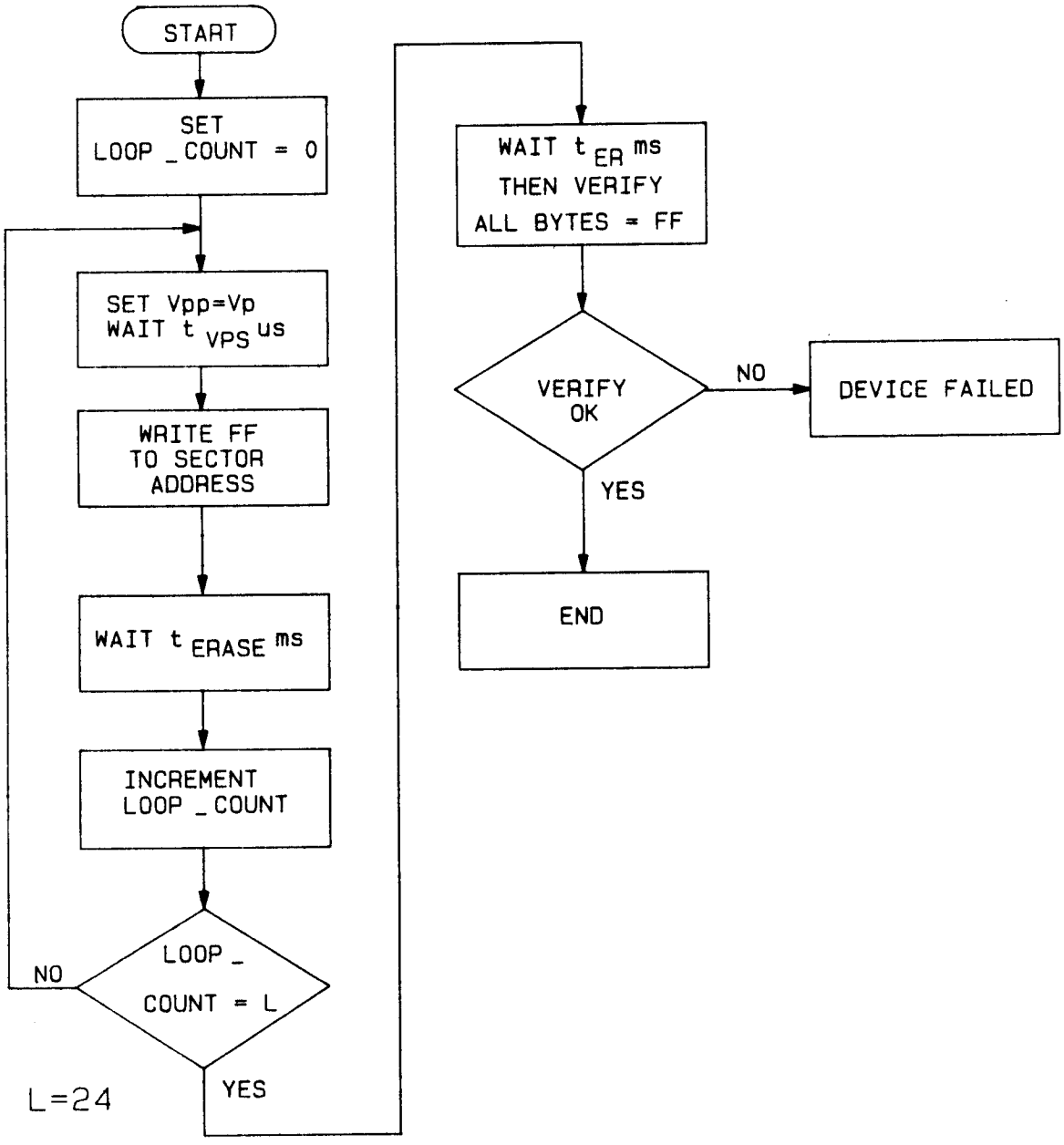


FIGURE 9. Sector clear algorithm.

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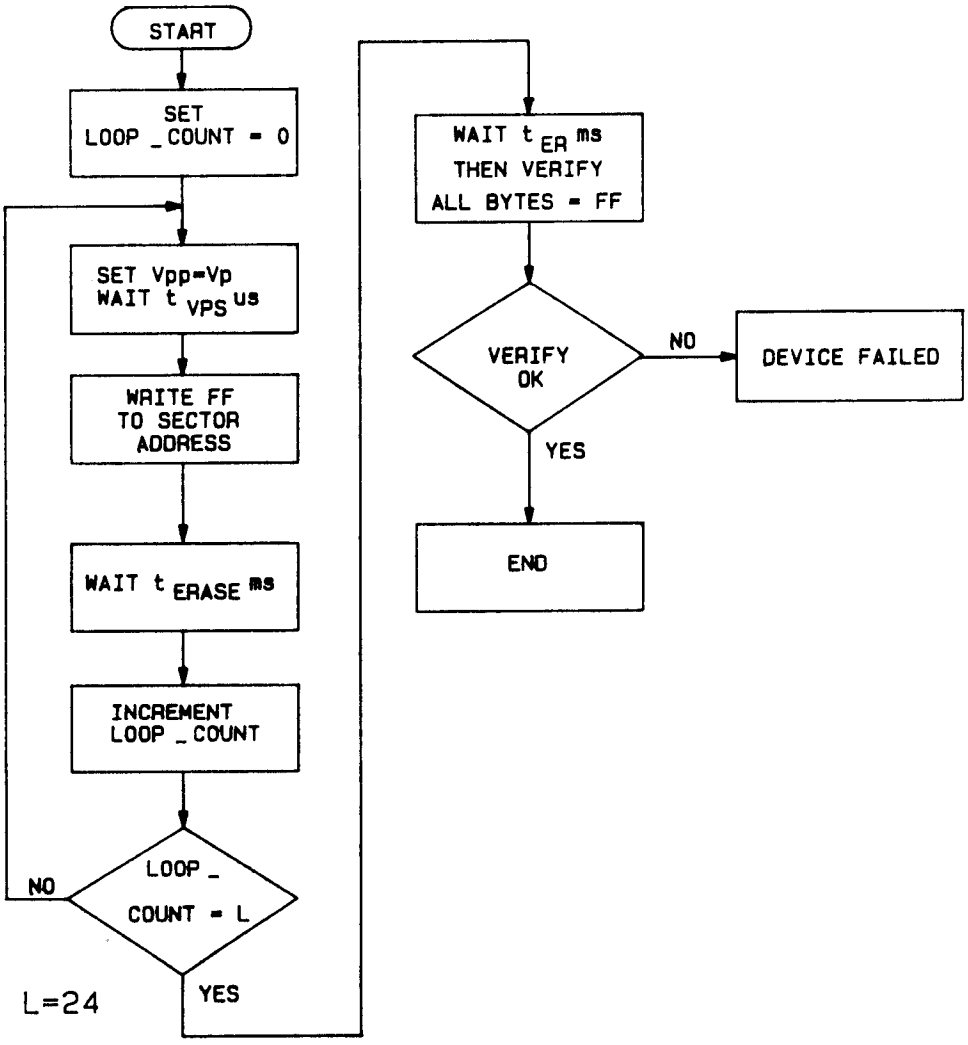


FIGURE 10. Chip clear algorithm.

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3.8.5 High voltage input protection flash EEPROMs: The V_{PP} pin must be at a high voltage for writing and clearing. There is an absolute maximum specification which must not be exceeded, even briefly, or permanent device damage may result. To minimize switching transients on this pin a minimum 0.1 µF, decoupling capacitor with good high frequency response should be connected from V_{PP} to ground. In addition, sufficient bulk capacitance should be provided to minimize V_{PP} voltage sag when a device goes from standby to a write or erase cycle.

4. QUALITY ASSURANCE PROVISION

4.1 Sampling and inspection. Samping and inspection procedures shall be in accordance with MIL-M-38510 and method 5005 of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Delete the sequence specified as initial (pre-burn-in) electrical parameters through interim (post-burn-in) parameters of method 5004 and substitute lines 1 through 6 of table II herein.

b. Prior to burn-in, the devices shall be programmed (see 3.8.3) with the data pattern shown on figure 4. The pattern shall be read before and after burn-in. Devices having bits not in the proper state after burn-in shall constitute a device failure and shall be included in the PDA calculation (see 4.2f).

c. Dynamic burn-in test (method 1015 of MIL-STD-883) test condition D or F using the circuit shown on figure 6 or equivalent. Test duration and temperature shall be in accordance with method 5004 of MIL-STD-883.

d. Static burn-in (method 1015 of MIL-STD-883, test condition A). T_A = +125°C minimum, test duration for the static test shall be 48 hours minimum for class S. The 48 hour burn-in shall be broken into two sequences of 24 hours each (static I and static II) followed by interim electrical measurements.

(1) All inputs and common inputs/outputs shall be connected through R_1 to V_{CC}, separate outputs may be open or connected to $V_{CC}/2$ ±0.5 V. Resistor R_1 is optional on both inputs and open outputs when V_{CC} is not applied, and required on outputs connected to $V_{CC}/2$. R_1 = 2 kΩ ±5%.

(2) V_{CC} = 5.75 V ±0.25 V.

(3) For static II burn-in change input connections to V_{SS}.

e. Interim and final electrical test parameters shall be as specified in table II. Interim electrical test parameters prior to burn-in shall be performed by the manufacturer. The following data patterns all "0's", all "1's", checkerboard, and checkerboard complement. Each temperature shall include, at a minimum, the programming of one data pattern. Subgroups 9, 10, and 11 shall be performed on devices containing a checkerboard data pattern or equivalent alternating bit data pattern.

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f. Percent defective allowable (PDA): The PDA for class B devices shall be as specified in MIL-M-38510. The PDA is specified as 5 percent for class B devices based on failures from group A, subgroups 1 and 7 after cooldown, at final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. All screening failures of group A, subgroups 1 and 7 after burn-in divided by the total number of devices submitted to burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA.

g. A data retention stress and margin test shall be included as part of the screening procedure prior to burn-in and shall consist of the following steps:

(1) All devices shall be programmed with a charge opposite the state that the cell would read in its equilibrium state (e.g., worst case pattern, see 3.8.3). Verify margin to be greater than 6.5 volts.

(2) After programming, perform a high temperature unbiased storage 72 hours at 150°C minimum. The storage time may be accelerated by using a higher temperature in accordance with the Arrhenius relationship and with apparent activation energy of .6 eV. The maximum storage temperature shall not exceed 200°C for assembled devices and 300°C for unassembled devices.

(3) Read the data retention pattern and test using subgroups 1, 7, and 9 (at the manufacturer's option, high temperature subgroups 2, 8A, and 10 or low temperature equivalent subgroups 3, 8B, and 11 may be used in lieu of subgroups 1, 7, and 9) after bake prior to burn-in. Verify margin to be greater than or equal to 6.0 volts.

h. After the completion of all screening, the devices shall be cleared and verified prior to delivery.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.3.1 Qualification extension. When authorized by the qualifying activity for qualification inspection, if a manufacturer qualifies to a faster device type which is manufactured identically to a slower device type on this specification, the slower device type may be part I qualified without further qualification testing. At the manufacturer's request, the slower device types will be added to the QPL.

4.3.2 Electrostatic discharge sensitivity qualification inspection. Electrostatic discharge sensitivity (ESDS) testing shall be performed in accordance with MIL-STD-883, method 3015. The option to categorize devices as ESD sensitive without performing the testing is not allowed. Only those device types that pass ESD testing at 1,000 volts or greater shall be considered as conforming to the requirements of this specification. ESD testing shall be measured only for initial qualification and after process or design changes which may affect ESDS classification.

4.4 Quality conformance inspections. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspection (see 4.4.1 through 4.4.4).

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4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Electrical test requirments shall be as specified in table II.

b. Subgroups 5 and 6 of table I of method 5005 of MIL-STD-883 shall be omitted.

c. Subgroup 4 (C_I and C_O measurements) shall be measured for initial qualification and after any process or design changes which may affect capacitance. Sample size is fifteen devices, with no failures, and all input and output terminals tested.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows:

a. Class S steady state life test circuits shall be conducted, using test condition D and the circuit described in 4.2b herein, or equivalent as approved by the qualifying activity.

b. Class S only, end-point electrical parameters shall be as specified in table II herein. Delta limits shall apply only to subgroup 5 of group B inspections and shall consist of tests specified in table III herein.

c. All class S devices selected for testing shall be written with the pattern shown on figure 3 (see 3.8.3).

4.4.3 Group C inspection. Group C inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows:

a. End point electrical tests shall be as specified in table II herein.

b. All devices requiring endpoint electrical testing shall be written with the pattern shown on figure 3.

c. Steady state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D or F in 4.5.2.

(2) Ambient temperature = +125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by method 1005 of MIL-STD-883.

(4) Read the pattern after burn-in and perform end-point electrical tests in accordance with table II herein for group C.

d. An endurance test, in accordance with method 1033 of MIL-STD-883, shall be added to group C inspection prior to performing the steady state life (see 4.4.3c) and extended data retention (see 4.4.3e) test. Cycling may be sector, byte, or chip from devices passing group A after the completion of the requirements of 4.2 herein. Initially two groups of devices shall be formed, cell 1 and cell 2. The following conditions shall be met:

(1) Cell 1 shall be cycled at -55°C and cell 2 shall be cycled at +85°C for a minimum of 100 cycles for device types 01, 03, 05, 07, 09, and for 1,000 cycles for device types 02, 04, 06, 08, and 10 (see 1.2.1).

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(2) Perform group A subgroups 1, 7, and 9 after cycling. Form new cells (cells 3 and 4) for steady state life and extended data retention. Cell 3 for steady state life test consists of one-half of the devices from cell 1 and one-half of the devices from cell 2. Cell 4 for extended data retention consists of the remaining devices from cells 1 and 2.

(3) The sample plans for cell 1, cell 2, cell 3, and cell 4 shall individually be the same as for group C1, as specified in method 5005 of MIL-STD-883.

e. Extended data retention shall consist of:

(1) Write all bits in each device with the data pattern representing the worst case data pattern.

(2) After cycling, perform a high temperature unbiased bake for 1,000 hours at 150°C minimum. The bake time may be accelerated by using a higher temperature in accordance with the Arrhenius relationship and with the apparent activation energy of .6 eV. The maximum bake temperature shall not exceed 175°C.

(3) Read the pattern after bake and perform endpoint electrical tests for table II herein for group C.

f. Cell, 1, cell 2, cell 3, and cell 4 must individually pass the specified sample plan.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883 and as follows:

a. End point electrical tests shall be as specified in table II herein.

b. All devices selected for electrical testing shall be written with the pattern shown on figure 6. After completion of all testing, the devices shall have the written pattern read, then be cleared and verified. When the use of electrical rejects is permitted, no writing or clearing or verification is required.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Voltages and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

4.5.2 Life test, burn-in, cooldown, and electrical test procedures. When devices are measured at +25°C following application of the steady state life or burn-in test condition, all devices shall be cooled to +35°C or within 10°C of power stable condition prior to removal of bias voltages/signals. Any electrical tests required shall first be performed at -55°C or +25°C prior to any required tests at +125°C.

4.5.3 Delta measurements. Delta measurements, as specified in table II, shall be made and recorded before and after the required burn-in screens and steady state life tests to determine delta compliance. The electrical parameters to be measured, with associated delta limits are listed in table III herein.

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TABLE III. Delta limits at +25°C (subgroup 1).

Parameter 1/	Device types
	All
V_{OL}	10 percent of specified value
V_{OH}	10 percent of specified value
I_{CC3}	10 percent of specified value
I_{LIH}, I_{LIL}	10 percent of specified value
I_{LOH}, I_{LOL}	10 percent of specified value

1/ The above parameters shall be recorded before and after the required burn-in and life tests to determine deltas.

4.5.4 Writing procedure. The waveforms and timing relationships shown on figures 4 and 6 and the conditions specified in table I and figure 7 shall be adhered to. Initially and after each chip clear (see 4.5.5), all bits are in the high state (output at V_{OH}).

4.5.4.1 Clear and write. Latches on address, data, and control inputs permit clearing and writing using normal microprocessor bus timing. Address inputs are latched on the falling edge of write enable or chip enable, whichever is later while data inputs are latched on the rising edge of write enable or chip enable whichever is earlier. The write enable input is noise protected; a pulse of less than 20 ns will not initiate a write or clear. In addition, chip enable, output enable and write enable must be in the proper state to initiate a write or clear. Timing diagrams depict write enable controlled writes; the timing also applies to chip enable controlled writes.

4.5.4.2 Byte write. A byte write is used to change any "1" in a byte to a "0"; to change a bit in a byte from a "0" to a "1", the byte must be erased first by either sector clear or chip clear. The organization of a group of bytes in flash EEPROMs is called a sector. The memory array is divided into 128 sectors of 512 bytes each. Individual bytes are written as part of a sector write operation. The write algorithm for either chip or sector write is detailed on figure 7. Sectors are written by applying a high voltage to the Vpp pin and writing individual non-FF bytes in sequential order. Each byte write is automatically latched on the chip, so that the user can do a normal microprocessor write cycle and then wait a minimum of t_{WC} ns for the self-timed write to complete. Each byte write incrementally writes bits to a zero. A write loop has been completed when all non-FF data for all desired sector locations has been written. Following each loop, a read verification is done. If any bytes do not verify, another write loop is performed. When all bytes read correctly, additional loops are performed to insure adequate bit cell margin. The total number of loops will vary by device

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and depends on temperature; low temperature reduces the number of loops required. For example, a typical (room temperature) loop count is 4. Sectors need not be written individually; the entire device or any combination of sectors can be written using the write algorithm. Because a partially written sector or one or more bytes in a sector must be changed, the contents of the sector must first be read into system RAM, the sector with new bytes cleared can then be added to the block of data in RAM and the sector rewritten using the sector write algorithm.

4.5.5 Clearing procedure. The waveforms and timing relationship shown on figures 4 and 6 and the conditions specified in table I and figure 7 shall be adhered to. Initially and after each chip clear, all bits are in the high state (output at V_{OH}).

4.5.5.1 Sector clear. Sector clear changes all bits in a sector of the array to a logical "1". It requires that the Vpp pin be brought to a high voltage and a write cycle performed. The sector to be cleared is defined by address inputs A9 through A15. The data inputs must be all ones to begin the clear. Following a write of "FF" the part will wait for time to abort to allow clearing before writing again. This allows recovery from unintentional sector clear. For example, in loading a sector of data a byte of "FF" was written. After the t_{ABORT} delay the sector clear will begin. The clear is detailed on figure 7. V_{PP} can be brought to any TTL level or left at high voltage after the clear.

4.5.5.2 Chip clear. Chip clear changes all bits in the memory to a logical "1". Refer to figure 7 for the chip clear algorithm. Vpp can be brought to any TTL level or left at high voltage after the clear.

4.5.6 Sector and chip clear. To reduce the sector and chip clear times, a software clear algorithm is used.

4.5.7 Read mode operation. The device is in the read mode whenever the $\overline{CE}$ and $\overline{OE}$ pins are at V_{IL} and the $\overline{WE}$ pin is at V_{IH}. The waveforms and timing relationships shown on figures 8, 9, and 10 and the test conditions and limits specified in table I shall be applied. Vpp may be at any level.

4.6 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of MIL-M-38510.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510. The devices covered by this specification require electrostatic protection.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Complete Part or Identifying Number (PIN) (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

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c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to procuring activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of MIL-STD-883, method 5003), corrective action and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special lead lengths or lead forming if applicable. These requirements shall not apply to direct purchase or direct shipment to the Government.

h. Requirements for "JAN" marking.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331 (including terms and symbols for device terminals), and as follows:

A_0-A_{12}	Address inputs used to address 1 of 2048/8 bit locations in static storage array.
DQ_0-DQ_7	Data I/O, 8 bit wide data bus.
I_{CC1}	Supply current (standby and active).
I_{CC2}	Supply current (TTL standby).
I_{CC3}	Supply current (CMOS standby).
I_{LIH}, I_{LIL}	Input leakage currents.
I_{LOH}, I_{LOL}	High impedance output leakage current.
V_{IL}	Logical low input voltage.
V_{IH}	Logical high input voltage.
V_{OL}	Logical low output voltage.
V_{OH}	Logical high output voltage.
C_I	Input capacitance.
C_O	Output capacitance.
t_{AVAV}	Cycle time from one read to next read.
t_{ELQV}	Chip enable access time.
t_{AVQV}	Address access time.
t_{OLQV}	Ouput enable access time.
V_{OH1}	Output level (TTL)
V_{OH2}	Output level (CMOS)

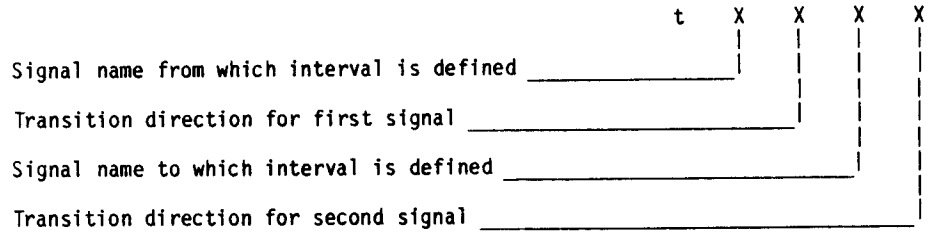
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t_{OHQX}	Output hold time.
t_{EHQZ}	$\overline{CE}$ to data float.
t_{OHQZ}	$\overline{OE}$ to data float.
t_{OLOX}	Output enable to output in low Z.
t_{WHWL}	Cycle time during $\overline{WE}$ write operation.
t_{EHEL}	Cycle time during CE write operation.
t_{AVWL}	Address to WE setup time.
t_{AVEL}	Address to CE setup time.
t_{WLAX}	Address hold time after WE low.
t_{ELAX}	Address hold time after CE low.
t_{ELWL}	Chip enable to WE setup time.
t_{WLEL}	Write enable to CE setup time.
t_{WHEH}	Chip enable hold time after WE high.
t_{EHWH}	Write enable hold time after CE high.
t_{ELEH}	Chip enable pulse width during write.
t_{OHWL}	Output enable to WE setup time.
t_{OHEL}	Output enable to CE setup time.
t_{WHOL}	Output enable hold time after WE high.
t_{EHOL}	Output enable hold time after CE high.
t_{WLWH}	Write enable pulse width during write.
t_{EHEL}	Minimum chip enable high time after write.
t_{DVWH}	Data in setup time before WE high.
t_{DVEH}	Data in setup time before CE high.
t_{WHDX}	Data hold time after WE high.
t_{EHDX}	Data hold time after CE high.
t_{ELWL}	CE set time (chip clear).
t_{WLWH}	$\overline{WE}$ pulse width (chip clear).
t_{WHOH}	$\overline{OE}$ hold time (chip clear).
V_P	Write/clear voltage.
V_{PRP}	Voltage during read.
I_{PS}	Vpp current.
I_{PR}	Current read mode.

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I_{pp}	Current byte write mode.
I_{PE}	Current clear mode.
t_{PVpWL}	Vpp setup time.
t_{WHPX}	Vpp hold time.
t_{WHEL}	Write recover time.
t_{OHWL}	$\overline{OE}$ setup time.
t_{WHEL}	Erase time.
t_{VPXEL}	Erase recover time.

6.4.1 Timing parameter abbreviations. All timing abbreviations use a lower case character with upper case character subscripts. The initial character is always t and is followed by four descriptors. These characters specify two signal points arranged by "from-to" sequence that define a timing interval. The two descriptors for each signal specify the signal name and the signal transition. (Note: There are exceptions for undefined signals).



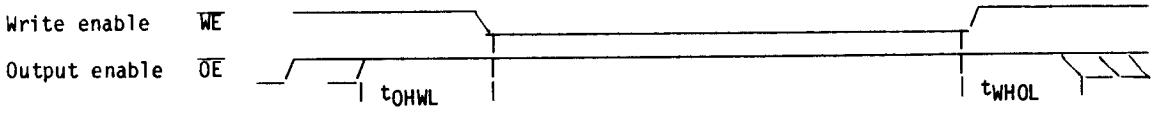
a. Signal definitions:

A = Address	D = Data in	Q = Data out
W = Write enable	E = Chip enable	O = Output enable
P = V_{CC}	R = Ready/busy	

b. Transition definitions:

H = Transition to high	L = Transition to low
V = Transition to valid	X = Transition to invalid
Z = Transition to high impedance	

Example:

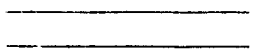
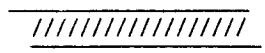
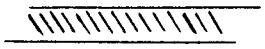
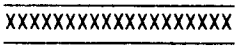


The example shows $\overline{OE}$ to $\overline{WE}$ setup time defined as t_{OHWL} and OE hold time after WE high defined as t_{WHOL}.

c. Timing limits. The table of timing values shows either a minimum or a maximum limit for each parameter. Input requirements are specified from the external system point of view. Thus, address setup time is shown as a minimum since the system must supply at least that much time (even though most devices do not require it). On the other hand, responses from the memory are specified from the device point of view. Thus the access time is shown as a maximum since the device will never provide data later than that time (even though most devices will supply data much sooner).

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d. Waveforms:

Waveform symbol	Input	Output
	Must be valid	Will be valid
	Change from low to high	Will change from low to high
	Change from high to low	Will change from high to low
	Do not care: any change permitted	Changing state unknown
	Not applicable	Change to high impedance

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits procured for Government logistic support will be procured to device class B (see 1.2.2), and lead material and finish C (see 3.3). Longer length and lead forming shall not affect the part number.

6.5 Handling. MOS devices shall be handled with certain precautions to avoid damage due to accumulation of static charge. Input protective devices have been designed in the chip to minimize the effect of this static buildup. However, the following handling practices are recommended:

a. Devices should be handled on benches with conductive and grounded surface.

b. Ground test equipment and tools.

c. Handling of devices by the leads should be avoided.

d. Store devices in conductive foam or carriers.

e. The use of plastic, rubber, or silk in the MOS area should be avoided.

f. Relative humidity should be maintained above 50 percent, if practical.

g. Operator should be grounded when handling device.

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6.6 Testing by inference. Testing by inference is the validation of the performance of a parameter by measurement of the correct performance of a dependent parameter or function.

Slash sheet number	Vendor CAGE number	Vendor similar part number
01BXX	61394	DJ48F512-300
01BYX	61394	LJ48F512-300
01BZX	61394	FJ48F512-300
02BXX	61394	DJ48F512K-300
02BYX	61394	LJ48F512K-300
02BZX	61394	FJ48F512K-300
03BXX	61394	DJ48F512-250
03BYX	61394	LJ48F512-250
03BZX	61394	FJ48F512-250
04BXX	61394	DJ48F512K-250
04BYX	61394	LJ48F512K-250
04BZX	61394	FJ48F512K-250
05BXX	61394	DJ48F512-200
05BYX	61394	LJ48F512-200
05BZX	61394	FJ48F512-200
06BXX	61394	DJ48F512K-200
06BYX	61394	LJ48F512K-200
06BZX	61394	FJ48F512K-200
07BXX	61394	DJ48F512-150
07BYX	61394	LJ48F512-150
07BZX	61394	FJ48F512-150
08BXX	61394	DJ48F512K-150
08BYX	61394	LJ48F512K-150
08BZX	61394	FJ48F512K-150
09BXX	61394	DJ48F512-120
09BYX	61394	LJ48F512-120
09BZX	61394	FJ48F512-120
10BXX	61394	DJ48F512K-120
10BYX	61394	LJ48F512K-120
10BZX	61394	FJ48F512K-120

CAGE number	Vendor name and address
61394	SEEQ Technology, Incorporated 1849 Fortune Drive San Jose, CA 95131

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-1172)

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